

Observations

1. Dataset Overview

- The dataset contains 891 passenger records and 12 columns.
- Age has 177 missing values.
- Cabin has 687 missing values.
- Embarked has 2 missing values.

2. Age Distribution

- Most passengers were between 20 and 40 years old.
- Very few passengers were older than 60.

3. Survival by Gender

- Female passengers had a significantly higher survival rate than males.
- Most male passengers did not survive.

4. Survival by Passenger Class

- First-class passengers had the highest survival rate.
- Third-class passengers had the lowest survival rate.

5. Correlation Analysis

- Fare and Survival show a positive relationship.
- Passenger Class and Survival show a negative relationship.
- Higher-class passengers generally paid higher fares.

Conclusion

The EDA revealed that gender, passenger class, and fare were important factors affecting survival. Female passengers and first-class passengers had significantly better survival chances. The dataset also contains missing values in Age, Cabin, and Embarked columns that should be handled before building predictive models.